there is a corresponding phase. Just as mountains and valleys determine the shape of the surface of the earth, the poles and zeros determine the shape of the s-plane. Unlike mountains and valleys, every pole and zero is exactly the same shape and size as every other pole and zero. The only unique characteristic a pole or zero has is its *location*. Poles and zeros are important because they provide a concise representation of the value at *any point in the s-plane*. That is, we can completely describe the characteristics of the system using only a *few parameters*. In the case of the *RLC* notch filter, we only need to specify four complex parameters to represent the system: $z_1$, $z_2$, $p_1$, $p_2$ (each consisting of a real and an imaginary part).

To better understand poles and zeros, imagine an ant crawling around the s-plane. At any particular location the ant happens to be (i.e., some value of $s$), there is a corresponding value of the transfer function, $H(s)$. This value is a complex number that can be expressed as the magnitude & phase, or as the real & imaginary parts. Now, let the ant carry us to one of the zeros in the s-plane. The value we measure for the real and imaginary parts will be *zero* at this location. This can be understood by examining the mathematical equation for $H(s)$ in Eq. 32-3. If the location, $s$, is equal to any of the zeros, one of the terms in the numerator will be zero. This makes the entire expression equal to zero, regardless of the other values.

Next, our ant journey takes us to one of the poles, where we again measure the value of the real and imaginary parts of $H(s)$. The measured value becomes larger and larger as we come close to the exact location of the pole (hence the name). This can also be understood from Eq. 32-3. If the location, $s$, is equal to any of the $p$'s, the denominator will be equal to zero, and the division by zero makes the entire expression infinity large.

Having explored the unique locations, our ant journey now moves randomly throughout the s-plane. The value of $H(s)$ at each location depends entirely on the positioning of the poles and the zeros, because there are *no* other types of features allowed in this strange terrain. If we are near a pole, the value will be large; if we are near a zero, the value will be small.

Equation 32-3 also describes how *multiple* poles and zeros interact to form the s-domain signal. Remember, subtracting two complex numbers provides the *distance* between them in the complex plane. For example, $(s - z_0)$ is the distance between the arbitrary location, $s$, and the zero located at $z_0$. Therefore, Eq. 32-3 specifies that the value at each location, $s$, is equal to the distance to all of the zeros *multiplied*, divided by the distance to all of the poles *multiplied*.

This brings us to the heart of this chapter: how the location of the poles & zeros provides a deeper understanding of the system's *frequency response*. The frequency response is equal to the values of $H(s)$ along the imaginary axis, signified by the dark line in the topographical plot of Fig. 32-7. Imagine our ant starting at the origin and crawling along this path. Near the origin, the distance to the zeros is approximately equal to the distance to the poles. This makes the numerator and denominator in Eq. 32-3